Year 10 Admission Assessment

Version 3.1, October 2026

Name	
Date	

Welcome

These activities help us get to know you.

Do your best.

It is fine if you cannot answer every question.

Your teacher will tell you when to start each part.

How to answer

Tick one box.

Which one is a fruit? a chair ☐ an apple ☑ a bus ☐

Write your answer in the box.

12 + 8 = **20**

Show your working. You can get marks for your working, even if your answer is wrong.

Working

4 × 25 = 100, 100 + 30 = 130

The parts of this paper

Part	What you do	Time
Listening	Listen to two talks. Answer 16 questions.	15 minutes
Reading	Read two texts. Answer the questions.	15 minutes
Writing	Write a description or an article.	10 minutes
Break		5 minutes
Maths	Fluency, word questions and applied problems.	25 minutes
Science	Questions with pictures, tables and graphs.	8 minutes
Interview	Talk with a teacher and read some sentences aloud.	15 minutes

You may use a pencil, a rubber and a ruler.

You may use a calculator **only** in Maths Section C.

Listening

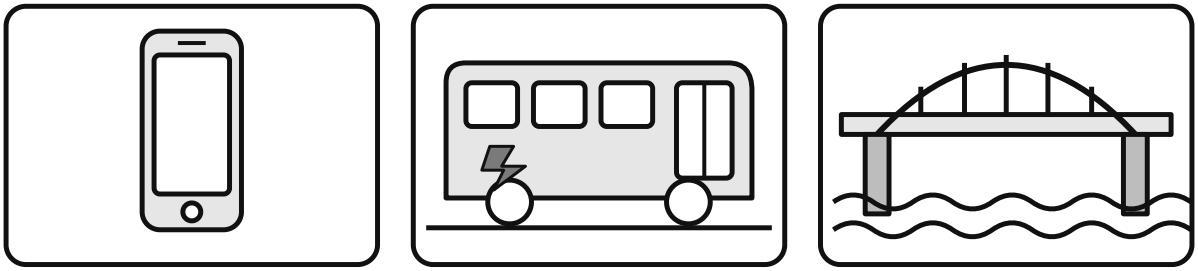
There are two parts: **A** and **B**.
Your teacher will read each part **two times**.
Listen carefully.
Before each part, you have one minute to read the questions.
You can answer while you listen, or after.

Part A

You will hear a short talk. **Answer** questions 1 to 8.

1 / 1

What does Dr Liu's company design?
Tick one picture.



A ☐ B ☐ C ☐

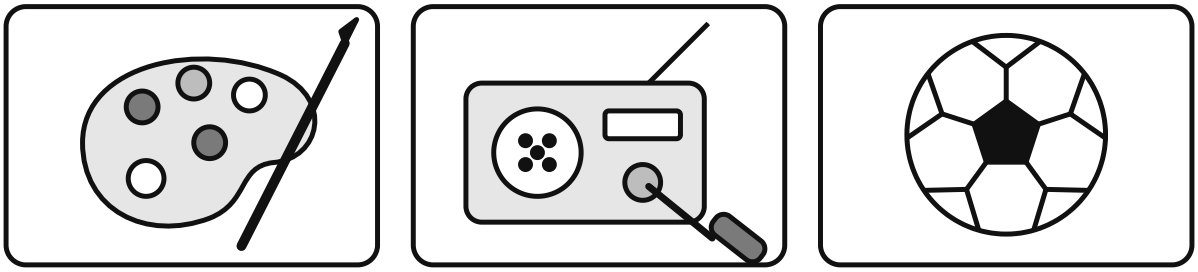
2 / 1

What was Dr Liu's favourite subject at school?
Tick one.

☐ Art ☐ Maths ☐ Physics

3 / 1

What did Dr Liu do at the weekends when she was young?
Tick one picture.



A ☐ B ☐ C ☐

4 / 1

How many years did she study at university? ☐

5 / 1

How many people are in her team? ☐

6 / 1

How far can the newest bus travel before it needs charging? ☐ km

7 / 1

Dr Liu says, "My parents weren't always happy about that!"
Why do you think her parents were not happy?

8 / 1

Why does Dr Liu tell the students that her first battery design failed every test?

Part B

You will hear a talk by **Professor Zhou Lan**. She is a scientist who studies the climate of cities.
Your teacher will read the talk **two times**.
You have one minute to read questions 9 to 16 first.

9 / 1

In the largest cities, on the hottest nights, how much warmer can the city centre be than the countryside?

[ ] degrees

10 and 11 / 2

Complete the notes. **Write** one to three words in each gap.

Example: Roads, concrete and dark roofs absorb the sun's energy in the day. They release it after [**sunset**].

Notes: why are cities hotter?

10 Plants cool the air. Water travels up from the soil, [ ] from the leaves and carries heat away.

11 Waste heat comes from traffic and from [ ].

12 / 1

A green roof has four layers.
Write 1, 2, 3 and 4 to show the order of the layers. 1 is the **bottom** layer.

Layer	Order
soil	[ ]
plants	[ ]
a waterproof layer	[ ]
a drainage layer	[ ]

13 / 1

Professor Zhou's team tested two roofs on a July afternoon.
What temperature did the surface of the **dark** roof reach? [ ] °C

14 / 1

Professor Zhou talks about a "vicious circle".

Explain how a green roof on a building could help to break this circle.

15 / 1

Builders often say that green roofs are too expensive.

What does Professor Zhou think about this? **Tick** one.

- ☐ She agrees that green roofs cost too much to build.
- ☐ She thinks they forget the cost of a city that is too hot.
- ☐ She thinks they should build white roofs instead.
- ☐ She thinks they do not know how green roofs are built.

16 / 1

What is the main purpose of Professor Zhou's talk? **Tick** one.

- ☐ To warn that cities will soon be too hot for people to live in.
- ☐ To explain a problem in cities and suggest how the school could help.
- ☐ To show that green roofs are cheaper than white roofs in the end.
- ☐ To report the results of her team's research in Hangzhou.

Reading

There are two parts: **A** and **B**.
Part B is harder than Part A.
Do as much as you can. It is fine if you do not finish.
You have 15 minutes. When you finish, wait for your teacher.

Part A

Read the menu. Then **answer** questions 1 to 7.

Lotus Café

Fresh noodles, rice and snacks, made every day

Opening hours
Monday to Friday: 7:30 to 20:00
Saturday and Sunday: 9:00 to 21:00

Noodles and rice

Beef noodle soup	spicy	¥32
Tomato and egg noodles	vegetarian	¥24
Chicken fried rice		¥28
Vegetable fried rice	vegetarian	¥22
Spicy tofu with rice	vegetarian, spicy	¥26

Snacks

Pork dumplings (8)		¥18
Spring rolls (4)	vegetarian	¥15

Drinks

Milk tea	¥14
Orange juice	¥16
Water	¥5

Key
vegetarian (no meat or fish)
spicy

Student deal
Any noodle dish
+ any drink = ¥35
Monday to Friday, 14:00 to 17:00
Show your student card.

Allergies?
Please tell our staff before you order.

1 / 1

What time does the café open on Sunday? _____

2 / 1

How much is the beef noodle soup? _____

3 / 2

Tick the **two** dishes that are vegetarian.

- ☐ Beef noodle soup
- ☐ Tomato and egg noodles
- ☐ Chicken fried rice
- ☐ Spring rolls
- ☐ Pork dumplings

4 / 1

Aisha is vegetarian. She does not like spicy food.

Write one dish from "Noodles and rice" that she can choose.

_____

5 / 1

When can you get the student deal? **Write** the days and the times.

_____

6 / 1

What must you show to get the student deal? _____

7 / 1

You have an allergy. What should you do?

_____

Part B

Read the article. Then **answer** questions 8 to 14.

Ten Minutes of Nothing

By Zhang Yiran, Year 11

When did you last do nothing at all? I do not mean sleeping, and I do not mean scrolling through videos. I mean sitting still, with no screen, no music and no plan, and simply letting your mind wander.

If you are like most teenagers I know, you probably cannot remember. The moment we feel even slightly bored (in a queue, on the bus, while a video is loading) our hands reach for our phones. Boredom feels like a problem, and our phones offer an instant solution.

I would like to suggest that boredom is not a problem at all. In fact, it may be one of the most useful feelings we have.

Psychologists describe boredom as a kind of signal. It tells us that what we are doing is not satisfying, and it pushes us to look for something better. When we fill every empty moment with a screen, we never hear that signal. We stay busy, but we are not really choosing what to do with our time.

There is also evidence that a wandering mind can be a creative one. In one study, a group of adults spent fifteen minutes copying numbers from a long list. Afterwards, they were asked to think of as many uses as possible for two plastic cups. They came up with more ideas than a group who had skipped the boring task. The researchers believe that boredom gives the mind time to make unexpected connections.

Of course, nobody enjoys being bored for hours, and I am not suggesting that we throw our phones into the nearest river. My suggestion is much smaller: try ten minutes of nothing each day. Leave your phone in your bag on the way home. Look out of the window. Notice where your thoughts go.

The first few minutes will feel strange, perhaps even uncomfortable. But you may be surprised by what your mind does when you finally give it some space.

8 / 1

What does the writer say most teenagers do when they feel bored?

9 / 1

What does "instant" mean in "our phones offer an instant solution"? **Tick** one.

☐ expensive ☐ immediate ☐ difficult ☐ boring

10 / 1

According to psychologists, what does boredom tell us?

11 / 1

In the study, what did the bored group do better than the other group?

12 / 2

Tick Fact or Opinion for each sentence.

Example: Hangzhou is a city in China. Fact ☑ Opinion ☐

Sentence	Fact	Opinion
In one study, a group of adults spent fifteen minutes copying numbers.	☐	☐
Boredom is not a problem at all.	☐	☐
The bored group came up with more ideas than the other group.	☐	☐
Boredom may be one of the most useful feelings we have.	☐	☐

13 / 1

"I am not suggesting that we throw our phones into the nearest river."
Why does the writer include this sentence?

14 / 1

The article begins with a question: "When did you last do nothing at all?"
Why do you think the writer begins in this way?

Writing

Choose one task: A or B.

Tick the box for the task you choose.

Plan your ideas in the plan box. Then **write** on the lines.

You have 10 minutes. Write as much as you can.

Task A: a description ☐

Describe a busy place you know well. For example: a market, a train station or a festival.

Help the reader to see, hear and smell the place.

Ideas: the people, the sounds, the smells, the colours, how the place makes you feel.

Task B: an article ☐

Some people say:

> Every student should spend one week each year volunteering in their community, for example in a library, a park or a home for older people.

Do you agree? **Write** an article for the school website. Give your opinion, with reasons and examples.

Plan

Notes and ideas (this part is not marked)

Your writing

Task A ☐ Task B ☐

Maths

You have 25 minutes. Do as much as you can.

Write your answer in the box. **Show** your working in the working box.

Your teacher can read any question to you.

Sections A and B: you may **not** use a calculator.

Section C: you **may** use a scientific calculator.

Section A: Fluency

No calculator

What the maths words mean

Simplify fully: write it as simply as you can, for example $\frac{2x + 4}{x + 2} = 2$

Surd form: keep the square root in your answer, for example $\sqrt{12} = 2\sqrt{3}$

Make *a* the subject: $b = a + 2$, so $a = b - 2$

f(x) = 2x + 1 means "double x, then add 1". So f(3) = 7.

fg(x) means "do g first, then do f". **$f^{-1}(x)$** is the **inverse** of f: it undoes f.

1 / 1

Write $\sqrt{48} - \frac{6}{\sqrt{3}}$ in the form $k\sqrt{3}$.

2 / 1

Work out $27^{-2/3}$. **Give** your answer as a fraction.

3 / 2

Simplify fully $\frac{2x^2 + 5x - 3}{x^2 - 9}$

Working

4 / 2

Make x **the subject** of $y = \dfrac{2x + 3}{x - 1}$

Working

$x =$

5 / 2

$f(x) = 3x - 1$ $g(x) = x^2 + 2$

(a) **Work out** $fg(2)$.

(b) **Find** $f^{-1}(x)$. $f^{-1}(x) =$

6 / 2

Solve $\dfrac{3}{x + 1} + \dfrac{2}{x} = 1$

Give your answers in surd form.

Working

$x =$ or $x =$

Section B: Word questions

No calculator

The questions in this section are in words only.
Read each question carefully. Decide what to do.

Write an equation, then solve it.
Three consecutive numbers add up to 87. Find the numbers.
(Consecutive numbers come one after another, like 5, 6 and 7.)

Let the smallest number be n. The numbers are n, $n + 1$ and $n + 2$.
Equation: $n + (n + 1) + (n + 2) = 87$
Solve: $3n + 3 = 87$, so $3n = 84$, so $n = 28$.
Answer: the numbers are 28, 29 and 30.

7 / 2

The price of a concert ticket goes up by 15%.
The new price is ¥230.
Work out the price before it went up.

Working

¥

8 / 2

At a museum, 3 adult tickets and 4 student tickets cost ¥290 altogether.
2 adult tickets and 5 student tickets cost ¥275 altogether.
Write two equations, then **solve** them.
Find the cost of one adult ticket and the cost of one student ticket.

Working

Adult ticket: ¥ Student ticket: ¥

9 / 2

A rectangle has a perimeter of 34 cm.

Its area is 60 cm^2.

Write an equation, then **solve** it.

Find the length and the width of the rectangle.

Working

Length: ☐ cm Width: ☐ cm

10 / 2

Pump A can fill an empty water tank in 4 hours.

Pump B can fill the same tank in 6 hours.

Both pumps are switched on at the same time.

How long do the two pumps take to fill the empty tank together?

Give your answer in hours and minutes.

Working

☐ hours ☐ minutes

Section C: Applied problems

You may use a scientific calculator

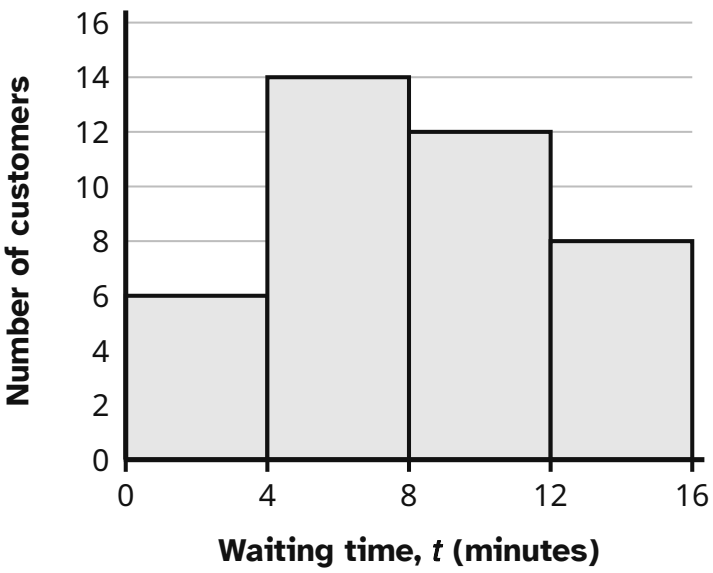
Show your working. You can get marks for your working, even if your answer is wrong.

11 / 2

A café records how long 40 customers wait for their food.
The chart shows the results.

Work out an estimate of the mean waiting time.

Working



minutes

12 / 2

A new laptop costs ¥6000.
Its value goes down by 12% every year.
After how many whole years is the laptop first worth **less than** ¥3000?
You can use the table.

Year	0	1	2	3	4	5	6	7
Value (¥)	6000	5280						

years

13

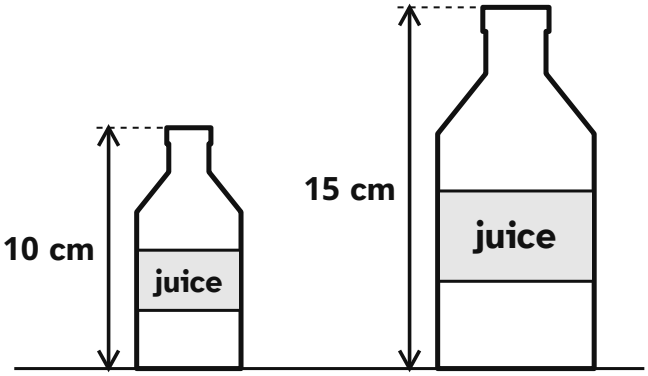
/ 2

These two bottles of juice are mathematically similar.

The small bottle is 10 cm tall.
It holds 200 ml. It costs ¥12.
The large bottle is 15 cm tall.
It costs ¥36.

Which bottle is better value? **Show** how you decide.

Not drawn accurately



Small: 200 ml, ¥12 **Large: ¥36**

Working

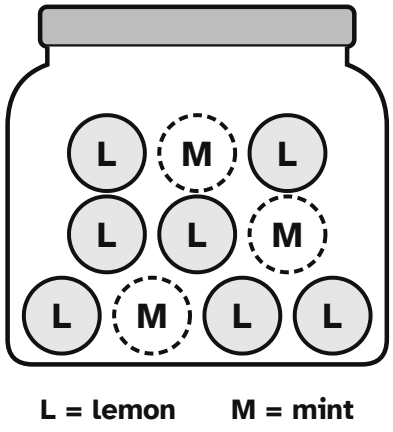
Tick one. Small bottle ☐ Large bottle ☐

14

/ 2

A jar has 10 sweets.
7 are lemon (L) and 3 are mint (M).
Jun takes a sweet at random and eats it.
Then he takes another sweet at random.

Work out the probability that the two sweets are the **same** flavour.



L = lemon **M = mint**

Working

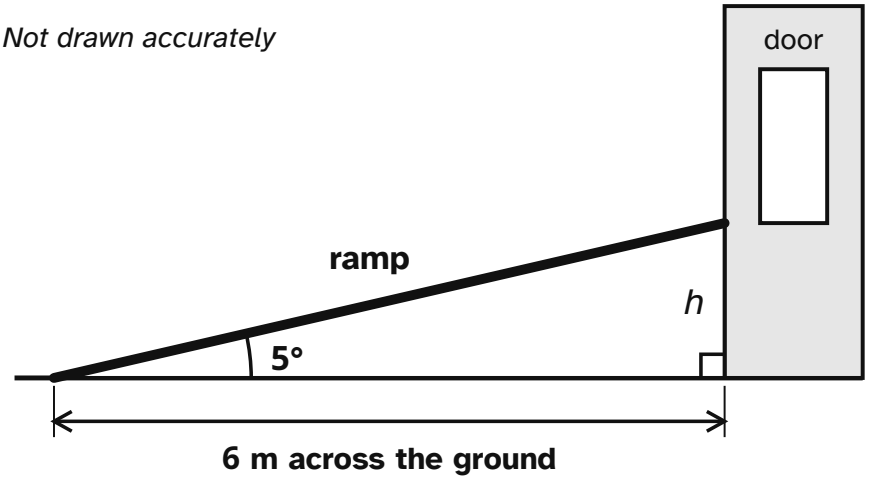
15

/ 2

This is a rule for wheelchair ramps:

> A ramp must not be steeper than **1 in 12**. For every 12 m across the ground, it can rise at most 1 m.

Not drawn accurately



This ramp goes 6 m across the ground. It makes an angle of 5° with the ground.
Does this ramp follow the rule? **Explain** your answer.

Working

Tick one. Yes, it follows the rule. ☐ No, it is too steep. ☐

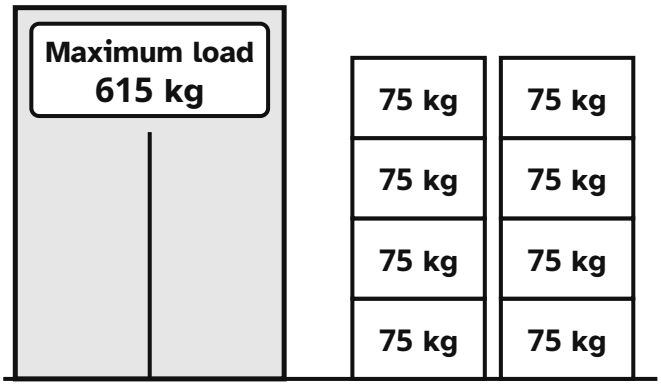
16

/ 2

A lift can carry a maximum load of 615 kg.
There are eight boxes.
Each box has a mass of 75 kg, correct to the nearest 5 kg.

Can the lift **always** carry all eight boxes at the same time? **Explain** your answer.

Working



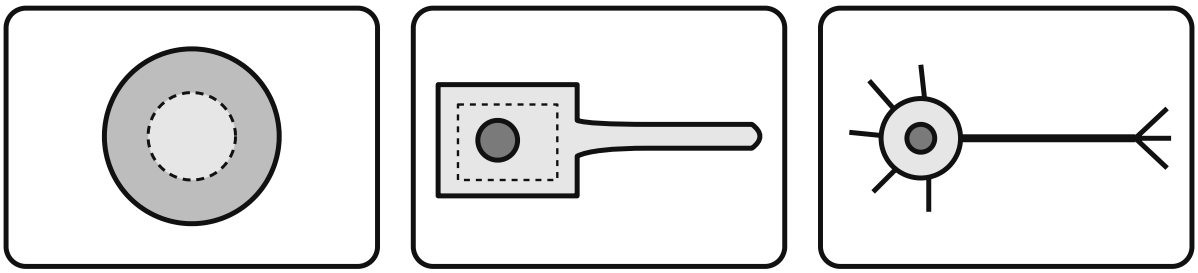
Tick one. Yes ☐ No ☐

Science

You have 8 minutes. Do as much as you can.
Your teacher can read any question to you.

1 / 1

These are three specialised cells.

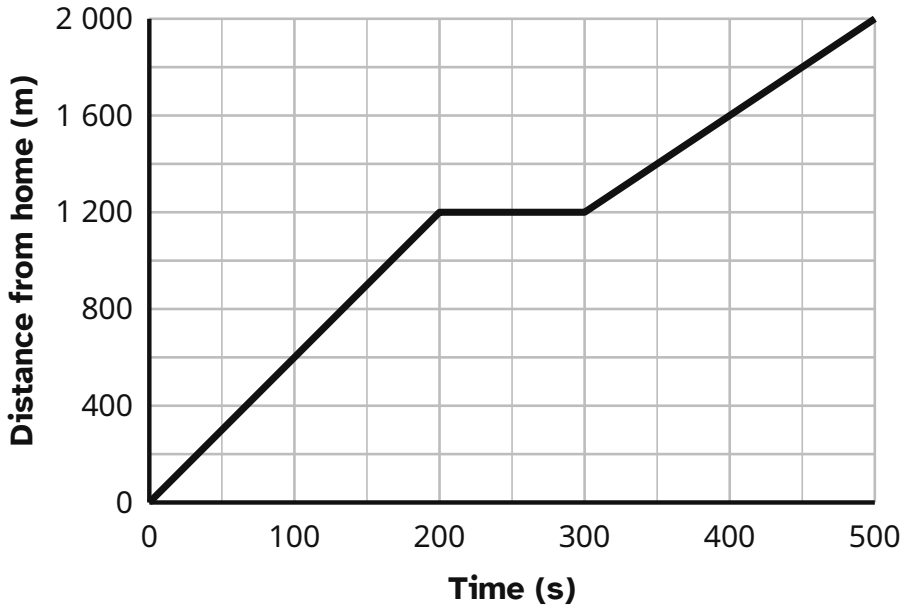


A **B** **C**

Which cell takes in water from the soil? **Write** the letter. ____

2 / 2

Jun cycles from home to school. This graph shows his journey.



(a) What is Jun doing between 200 s and 300 s? ____

(b) **Work out** Jun's speed in the first 200 s. Use: speed = distance ÷ time.

____ m/s

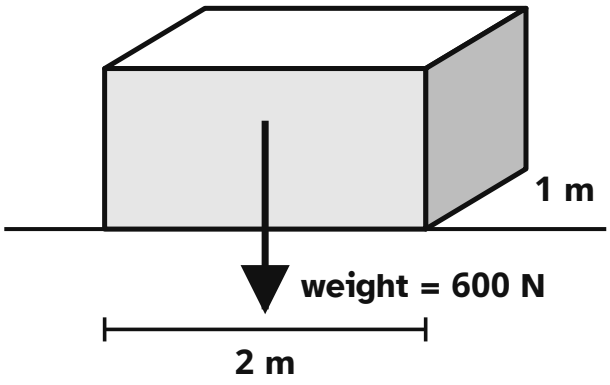
3 / 1

A box weighs 600 N.
Its base is 2 m long and 1 m wide.

Work out the pressure of the box on the floor.

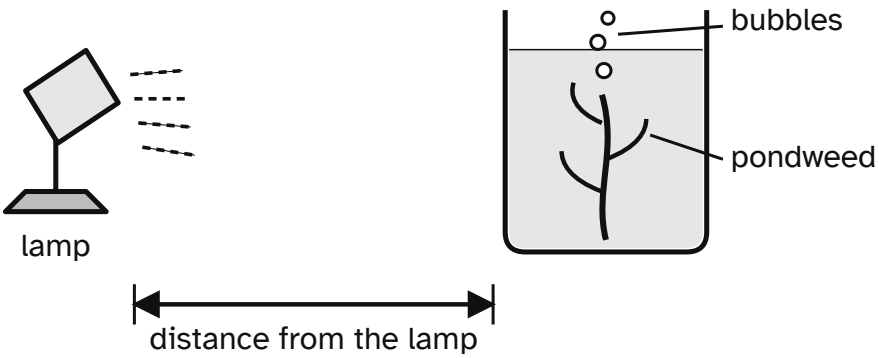
Use: pressure = force ÷ area.

_____ N/m^2



4 / 3

Sofia investigates how light affects photosynthesis. She puts a lamp at different distances from some pondweed. She counts the bubbles of gas in one minute.



Distance from the lamp (cm)	**10**	**20**	**30**	**40**
Number of bubbles in one minute	52	31	17	9

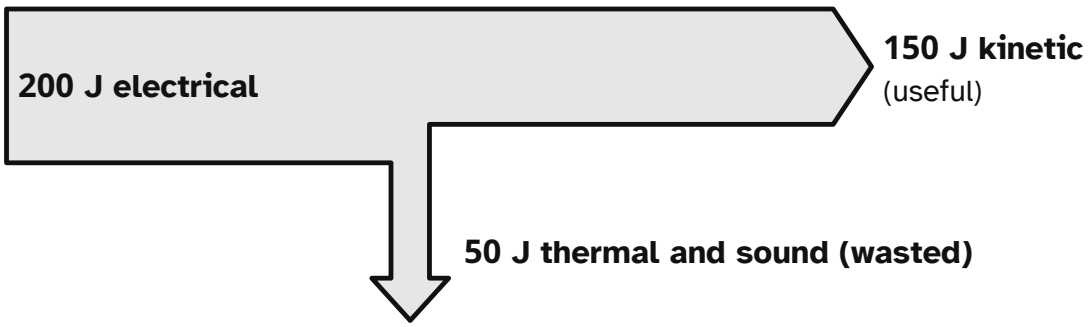
(a) **Describe** the pattern in the results.

(b) **Write** one variable Sofia must keep the same.

(c) Sofia counted the bubbles only once at each distance. **Suggest** one way to improve her method.

5 / 1

This diagram shows the energy transfers in an electric fan.

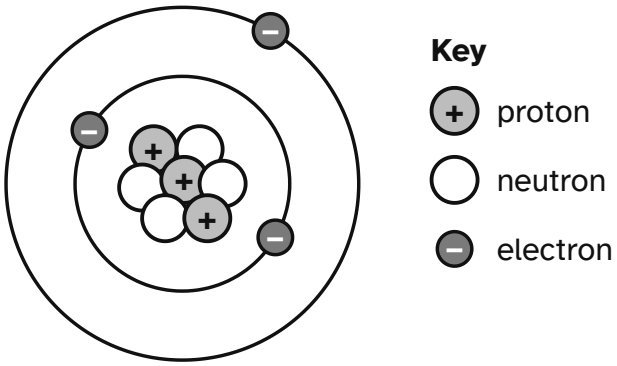


Work out the efficiency of the fan. Use: efficiency = useful energy out ÷ total energy in.

6 / 1

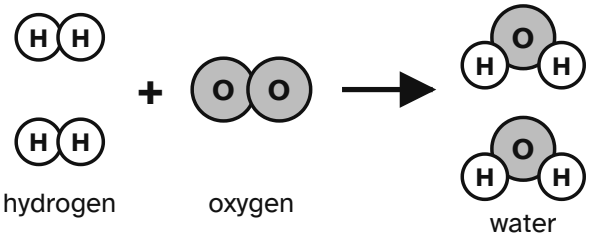
This diagram shows an atom.

What is the **mass number** of this atom?



7 / 1

Hydrogen reacts with oxygen to make water.



Write the missing number to balance the equation.

$$\square\ H_2 + O_2 \rightarrow 2H_2O$$

Thank you. You have finished. Please wait for your teacher.